COM3021

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

COMPUTER SCIENCE

Examination, May 2024

Data Science at Scale

Module Leader: Dr. Hugo Barbosa

Duration: 2 HOURS

- All questions are compulsory.
- The word estimates are provided as a guidance for the expected length of an effective and concise answer to the questions.

The marks for this module are calculated from 70% of the percentage mark for this paper plus 30% of the percentage mark for associated coursework.

No electronic calculators of any sort are to be used during the course of this examination.

This is a CLOSED BOOK examination.

Question 1

(a) Similar to a relational database in the sense that the data is modelled in tables, rows and columns. One of the key differences, however, is that in this approach the data is grouped in columns rather than rows.

Q: *What is being described in the previous paragraph?*

(1 mark)

(b) It enables an abstraction at the operating system level, allowing multiple applications to share binaries while remaining isolated from one another.

Q: *What is being described in the previous paragraph?*

(1 mark)

(c) A document is usually stored in a single continuous string like JSON, XML or BSON with the advantage that all the data for a given object is stored together.

Q: *What is being described in the previous paragraph?*

(1 mark)

(d) Reconstruct the matrix using the compressed sparse row (CSR) representation below and calculate its sparsity.

$$\begin{aligned} V &= [1,2,3,1,1,1,3,2,2,1] \\ ROW\_INDEX &= [0,2,3,5,7,10] \\ COL\_INDEX &= [0,2,1,2,4,0,2,1,3,4] \end{aligned}$$

(7 marks)

(Total 10 marks)

Question 2

You are the data science lead for a social media platform called PikPok in which users post short videos to their followers. Your team is responsible for the design of the machine learning models behind the PikPok platform.

- On PikPok, users can **scroll** on their Home page and watch content posted by the creators they follow.
- Users can also interact with the content through "likes" and comments.
- Moreover, users will also see on their Home page videos recommended by predictive models (based on their preferences/interactions).
- On PikPok users will post videos along with descriptive *hashtags* (e.g., #food, #funny, #outdoors)

(a) You are designing the content recommendation system of PikPok. One of the parts of the PikPok's content recommendation system (called PikPokIT) is a Machine Learning model that is continuously updated using millions of pieces of content and activities generated by all users of the platform every day.

Q: *Regarding the training process of the PikPokIT models behind its recommendation system, would you use full batch, mini-batch, online learning or a combination of them? Describe and justify your choice. (A complete answer will have between 60 and 100 words.*

(20 marks)

(b) One of the goals of PikPokIT is to maximise user engagement (i.e., how much users are interacting with the content they are watching). Your system will use the past and current users behaviours (**likes, video watching history, trending videos**), combined with ***hashtags*** to recommend content a user is more likely to interact with.

Q: *What data processing approach will you use to* ***suggest*** *the next videos to be shown* ***while a user is scrolling through their Home page****? Justify your choice highlighting the advantages of your strategy for PikPok in comparison with other competing approaches.* (a complete answer will have between 20 and 40 words)

(10 marks)

(c) During a meeting with your team, one of the data scientists in the team informed you of a problem in PikPokIT and suggested the usage of ADWIN as a solution.

Q: *What problem was being discussed? How was the problem manifesting itself? How can ADWIN help with that?*

(15 marks)

(Total 45 marks)

Question 3

(a) In the design of distributed data systems like cloud-based storage solutions, **explain** (*1*) the data storage strategies employed to ensure data **availability** across multiple nodes or data centres, and (*2*) how these strategies impact the overall system's **scalability**. *(a complete answer will have between 50 and 70 words)*

(15 marks)

(Total 15 marks)

Question 4

(a) In the context of distributed data systems, explain in details the four A.C.I.D. properties. (a complete answer will have between 60 and 100 words)

(15 marks)

(b) Explain the **linearizability** requirement, describing its trade-offs and how they are related to the trade-off underpinning the CAP theorem. (a complete answer will have between 50 and 90 words)

(12 marks)

(c) In the context of a **global social media platform with millions of users**, **explain and justify** how would you approach the CAP trade-off. (a complete answer will have between 10 and 20 words)

(3 marks)

(Total 30 marks)